8.4 Practice: word problems, step by step

← Unit overview · Answers

Every word problem gets the same five steps. Keep your card next to you:

> **1. Read** (underline the question) → **2. Picture** (draw it) → **3. Plan** (what first, what next?) → **4. Calculate** → **5. Check** (does it make sense? right unit?)

There is no clock. A drawing is always allowed and always smart.

See it done: six worked examples

Read these with your parent first. Notice that every one uses the same five steps.

Example 1: The school trip

98 children and 8 teachers go on a school trip. A bus has 50 seats. How many buses are needed?

1. **Read.** Question: how many buses?
2. **Picture.** Everyone needs a seat: 98 + 8 people. Buses of 50.
3. **Plan.** First count all the people, then see how many 50s fit.
4. **Calculate.** 98 + 8 = 106. 2 buses = 100 seats, not enough. 3 buses = 150 seats.
5. **Check.** 106 : 50 = 2,12. You can't book 0,12 of a bus, so round **up**. **3 buses.**

Example 2: Jacket in the sale

A jacket costs €48. There is 25% korting. You pay with a €50 note. How much change do you get?

1. **Read.** Question: how much change?
2. **Picture.** A strip: 100% = €48, 25% = a quarter.
3. **Plan.** First the discount, then the new price, then the change.
4. **Calculate.** 25% of €48 = €12. €48 − €12 = €36. €50 − €36 = €14.
5. **Check.** Change should be less than €50 and more than €2 (the change without korting). **€14.** ✓

Example 3: Home after the film

The film starts at 19:15 and lasts 2 hours 10 minutes. It takes 20 minutes to cycle home. What time are you home?

1. **Read.** Question: what time home?
2. **Picture.** A time line: 19:15 → film → end → cycle → home.
3. **Plan.** Add the film, then add the cycling.
4. **Calculate.** 19:15 + 2 h = 21:15; + 10 min = 21:25; + 20 min = 21:45.
5. **Check.** The answer is a time, not a number of minutes. **21:45.** ✓

Example 4: More pancakes

You need 3 eggs for 12 pancakes. You want to make 20 pancakes. How many eggs?

1. **Read.** Question: how many eggs for 20?
2. **Picture.** A ratio table.

pancakes	12	4	20
eggs	3	1	5

3. **Plan.** Find “1 egg makes how many pancakes”, then go to 20.
4. **Calculate.** 12 : 3 = 4 pancakes per egg. 20 : 4 = 5 eggs.
5. **Check.** 20 is a bit less than double 12, so a bit less than 6 eggs. **5 eggs.** ✓

Example 5: The extra information

Mila is 11 years old. She has €25 saved. She buys 2 books of €7,95 each. How much does she have left?

1. **Read.** Question: how much left?
2. **Picture.** Cross out “11 years old”: it doesn’t help.
3. **Plan.** Cost of 2 books, then take that from €25.
4. **Calculate.** 2 × €7,95 = €15,90 (think: 2 × €8 = €16, minus 10 cents). €25 − €15,90 = €9,10.
5. **Check.** About €25 − €16 = €9. **€9,10.** ✓

Example 6: Cycling on the map

On a map with scale 1 : 50 000 your route is 9 cm. You cycle at 15 km/u. How many minutes does the ride take?

1. **Read.** Question: how many minutes?
2. **Picture.** Map → real distance → time.
3. **Plan.** First the real distance, then the time with a ratio table.
4. **Calculate.** 1 cm = 500 m, so 9 cm = 4 500 m = 4,5 km. 15 km in 60 min → 1,5 km in 6 min → 4,5 km in 18 min.
5. **Check.** 4,5 km is less than a third of 15 km, so less than 20 minutes. **18 minutes.** ✓

Set A: Just plan it (Warm-up, no calculating needed)

For each problem, choose the calculation that fits. You don’t have to work it out (but you may, if you want).

1. Ahmed buys 3 cinema tickets of €7,50 each. How much does he pay?
   (a) 3 + 7,50 (b) 3 × 7,50 (c) 7,50 − 3
2. 96 children go to a sports day in buses with 32 seats. How many buses are full?
   (a) 96 − 32 (b) 96 × 32 (c) 96 : 32
3. Emma has €15 and gets €8 for her birthday. How much does she have now?
   (a) 15 + 8 (b) 15 − 8 (c) 15 × 8
4. A rope is 12 m long. You cut off 2,5 m. How long is the rest?
   (a) 12 + 2,5 (b) 12 − 2,5 (c) 12 : 2,5
5. A box of 24 chocolates is shared fairly among 6 people. How many each?
   (a) 24 − 6 (b) 24 × 6 (c) 24 : 6
6. 1 kg of cheese costs €12. What does 0,5 kg cost?
   (a) 12 : 2 (b) 0,5 × 12 (c) 12 + 0,5
7. A train leaves at 10:40 and arrives at 12:05. How long is the journey?
   (a) 10:40 + 12:05 (b) count on from 10:40 to 12:05 (c) 12:05 × 10:40
8. In a hall there are 4 rows of 7 chairs. How many chairs?
   (a) 4 + 7 (b) 4 × 7 (c) 7 − 4

Talk about it: Number 6 has a surprise. What is it?

Set B: One step (Warm-up → Building)

1. Sanne has €12. She buys a book for €7. How much does she have left?
2. 4 friends share 36 stroopwafels fairly. How many does each friend get?
3. A train ticket costs €8 per person. How much for 3 people?
4. A bus has 50 seats. 28 children get on. How many seats are empty?
5. A film starts at 14:30 and lasts 1 hour 45 minutes. What time does it end?
6. Juice boxes come in packs of 6. You need 30 juice boxes. How many packs?
7. Jan cycles 4 km to school and 4 km back home each day. How far does he cycle in one day?
8. Apples cost €2 per kilo. What do 3 kilos cost?

Talk about it: Which problem did you draw a picture for?

Set C: Two steps (Building)

1. You buy 3 ice creams of €2,50 each and pay with €10. How much change do you get?
2. A class of 27 children is split into groups of 4. Every child must be in a group. How many groups are needed?
3. There are 3 pizzas, each cut into 8 slices. 5 friends share them fairly. How many slices does each friend get, and how many are left over?
4. At a theme park a ticket costs €45 for an adult and €40 for a child. What do 2 adults and 2 children pay?
5. Lisa saves €3,50 a week. How many weeks does she need to save €28 for a game?
6. At the bakery you buy 4 croissants of €1,25 and one loaf of bread for €2,80. What do you pay?
7. A bottle holds 1,5 litres. How many glasses of 250 ml can you pour from it?
8. You leave at 9:45. The journey takes 2 hours and 35 minutes. What time do you arrive?
9. It rained 12 mm on Monday, 8 mm on Tuesday and 16 mm on Wednesday. What was the average (*gemiddelde*) per day?
10. There are 10 cookies in the tin. You eat 3. How many are left?

Talk about it: In number 2, why isn't the answer 6,75?

Set D: Several steps, and extra information (Stretch)

Cross out anything you don't need before you start.

1. A school has 8 classes of 25 children. Three quarters (¾) of all the children go on the school trip. How many children go?
2. Those children go by bus. A bus holds 50 children. How many buses are needed?
3. One litre of paint covers 8 m² of wall. You want to paint a wall that is 3 m high and 4 m wide. How many litres do you need? Paint is sold in 1-litre tins. How many tins do you buy?
4. A phone costs €240. You pay in 6 equal monthly amounts. How much per month?
5. Tom is 12 years old. He cycles 3 km to the swimming pool at 15 km/u. How many minutes does the ride take?
6. A camping pitch costs €22,50 per night. You stay 4 nights. You also pay €5 tourist tax (*toeristenbelasting*) once. What is the total?
7. At the market a stall-holder buys 50 apples at €0,20 each and sells them all at €0,35 each. How much profit (*winst*) does he make?

8. 600 children voted for a new playground. 45% voted for a climbing net. How many children is that?
9. A recipe for 4 people uses 3 eggs. How many eggs for 12 people?
10. There are 5 boxes with 12 pencils in each box. How many pencils?

Talk about it: Which piece of information in number 5 did you cross out?

Set E: Tables and timetables (Building → finish on a win)

Bus 15 timetable (*dienstregeling*)

Stop			
Station	8:05	8:35	9:05
Markt	8:12	8:42	9:12
Ziekenhuis	8:20	8:50	9:20
Strand	8:41	9:11	9:41

1. How often does bus 15 leave from the station?
2. How long does the ride take from Station to Strand?
3. You get on at Markt. You must be at the Ziekenhuis (hospital) by 9:00. Which is the latest bus you can take from Markt?
4. How long is the ride from Markt to Strand?

Visitors to the swimming pool (*zwembad*) in one week

Day	Mon	Tue	Wed	Thu	Fri
Visitors	120	95	210	80	145

5. On which day were there the most visitors?
6. How many visitors were there in total from Monday to Friday?
7. What was the average number of visitors per day?
8. How many more visitors were there on Wednesday than on Thursday?
9. How many visitors were there on Monday?

Talk about it: Why do you think Wednesday was so busy? (There's no right answer; Dutch primary schools often have a free Wednesday afternoon.)

Puzzle corner (optional, just for fun)

1. A frog is at the bottom of a 10 m well. Every day it climbs 3 m, every night it slides back 2 m. On which day does it get out?
2. A farmer has chickens and cows. Together they have 10 heads and 28 legs. How many chickens and how many cows?
3. Make up a word problem whose answer is "€4,50". Make it as strange as you like.